

Jiachen (Amber) Liu

amber@ara-commons.com | (734) 881 4120 | <https://amberljc.github.io/> | Palo Alto, CA

EDUCATION

University of Michigan (U of M)

Ann Arbor, MI

Ph.D. in Computer Science; Advised by Prof. Mosharaf Chowdhury

Aug 2020 - May 2025

- **Dissertation:** User-Centric Machine Learning Systems
- **Honors:** Ph.D. Student Fellowship (U of M). ML and Systems Rising Stars (Program of 2023)

B.S.E. in Data Science, Minor in Mathematics

Sep 2018 - May 2020

- **Honors:** Dean's List (2018), Dean's List (2019), 2019 University Honors

Shanghai Jiao Tong University (SJTU)

Shanghai, China

B.S. in Electrical Computer Engineering

Sep 2016 - Aug 2020

- **Honors:** Shanghai Outstanding Graduate (Top 5%), Dean's List (Top 5%), Undergraduate Scholarship (Top 10%)

Massachusetts Institute of Technology

Cambridge, MA

Visiting researcher at EECS CSAIL, Advised by Prof. Samuel Madden

May 2019 - Jan 2020

PUBLICATION

(* = equal contribution)

1. Agentic Auto-Research is Fuzz Testing. [Arxiv 2026](#).

Yifeng He, Jicheng Wang, Yinzhe Zhao, **Jiachen Liu**, Hao Chen.

2. The Last Human-Written Paper: Agent-Native Research Artifacts. [Arxiv 2026](#). (ICML & CAIS Workshop Spotlight)

Jiachen Liu, Jiaxin Pei, Jintao Huang, Chenglei Si, Ao Qu, Xiangru Tang, Runyu Lu, Lichang Chen, Xiaoyan Bai, Haizhong Zheng, et al. (37 authors), Alex Pentland, Ang Chen, Mosharaf Chowdhury, Zechen Zhang.

3. Sci-Reasoning: A Dataset Decoding AI Innovation Patterns. [Arxiv 2026](#).

Jiachen Liu*, Maestro Harmon*, Zechen Zhang.

4. EXP-Bench: Can AI Conduct AI Research Experiments? [ICLR 2026](#).

Patrick Kon*, **Jiachen Liu***, Xinyi Zhu, Qiuyi Ding, Jingjia Peng, Jiarong Xing, Yibo Huang, Yiming Qiu, Jayanth Srinivasa, Myungjin Lee, Mosharaf Chowdhury, Matei Zaharia, Ang Chen.

5. Humanity's Last Exam. [Nature 2026](#).

Long Phan, Alice Gatti, ..., **Jiachen Liu**, ..., Summer Yue, Dan Hendrycks (Scale AI & CAIS collaborative benchmark).

6. User-Centric Machine Learning Systems. [PhD Dissertation 2025](#).

Jiachen Liu

7. The ML.ENERGY Benchmark: Toward Automated Inference Energy Measurement and Optimization. [NeurIPS 2025](#). (Spotlight)

Jae-Won Chung, **Jiachen Liu**, Jeff Ma, Ruofan Wu, Oh Jun Kweon, Yuxuan Xia, Zhiyu Wu, Mosharaf Chowdhury

8. Curie: Toward Rigorous and Automated Scientific Experimentation with AI Agents. [Arxiv 2025](#).

Patrick Kon*, **Jiachen Liu***, Qiuyi Ding, Yiming Qiu, Zhenning Yang, Yibo Huang, Jayanth Srinivasa, Myungjin Lee, Mosharaf Chowdhury, Ang Chen.

9. Venn: Resource Management Across Federated Learning Jobs. [MLSys 2025](#).

Jiachen Liu, Fan Lai, Ding Ding, Yiwen Zhang, Mosharaf Chowdhury.

10. Evaluation Framework for AI Systems in "the Wild". [Arxiv 2025](#).

Sarah Jabbour, Trenton Chang, Anindya Das Antar, Joseph Peper, Insu Jang, **Jiachen Liu**, Jae-Won Chung, Shiqi He, Michael Wellman, Bryan Goodman, Elizabeth Bondi-Kelly, Kevin Samy, Rada Mihalcea, Mosharaf Chowdhury, David Jurgens, Lu Wang

11. IaC-Eval: A code generation benchmark for Infrastructure-as-Code programs. In NeurIPS 2024.
Patrick Kon, **Jiachen Liu**, Yiming Qiu, Weijun Fan, Ting He, Lei Lin, Haoran Zhang, Owen M. Park, George Sajan Elengikal, Yuxin Kang, Ang Chen, Mosharaf Chowdhury, Myungjin Lee, Xinyu Wang.
12. Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services. Arxiv 2024.
Jiachen Liu, Zhiyu Wu, Jae-Won Chung, Fan Lai, Myungjin Lee, Mosharaf Chowdhury.
13. Efficient Large Language Models: A Survey. In TMLR 2024.
Zhongwei Wan, Xin Wang, Che Liu, Samiul Alam, Yu Zheng, **Jiachen Liu**, Zhongnan Qu, Shen Yan, Yi Zhu, Quanlu Zhang, Mosharaf Chowdhury, Mi Zhang.
14. FedTrans: Efficient Federated Learning for Heterogeneous Clients via Model Transformation. In MLSys 2024.
Yuxuan Zhu, **Jiachen Liu**, Fan Lai, Mosharaf Chowdhury.
15. Auxo: Efficient Federated Learning via Scalable Cohort Identification. In SoCC 2023.
Jiachen Liu, Fan Lai, Yinwei Dai, Aditya Akella, Harsha Madhyastha, Mosharaf Chowdhury.
16. FedScale: Benchmarking Model and System Performance of Federated Learning. In ICML 2022.
Fan Lai, Yinwei Dai, Sanjay Singapuram, **Jiachen Liu**, Xiangfeng Zhu, Harsha Madhyastha, Mosharaf Chowdhury.
17. Fluid: A Generic Resource-aware Hyperparameter Tuning Execution Engine. MLSys 2021.
Peifeng Yu*, **Jiachen Liu***, Mosharaf Chowdhury.

WORK EXPERIENCE

ARA Commons, Inc. (ARA Lab), Founder & CEO	Palo Alto, CA
Area: Executable and Verifiable Research Infrastructure. Backed by DCVC.	May 2026 - Now
Meta Superintelligence Labs, Research Scientist	Menlo Park, CA
Area: LLM Pre- & Post-training Systems (Llama).	Jun 2025 - Apr 2026
Meta (AI & Systems Co-Design), Research Scientist Intern / Part-time	Menlo Park, CA
Area: Cross-Regional Llama Pre-training Systems.	May 2024 - Dec 2024
Apple, PhD Intern	New York, NY
Area: Private Machine Learning Framework.	May 2022 - Aug 2022
University of Michigan (SymbioticLab), Graduate Research Assistant	Ann Arbor, MI
Area: LLM Serving & Training Systems; Federated Learning.	Aug 2020 - May 2025

TEACHING EXPERIENCE

Graduate Student Instructor , EECS 598 Systems for GenAI (U of M)	2024 Winter
Teaching Assistant , EECS 484 Database Systems (U of M)	2019 Fall, 2020 Winter, 2020 Summer

SELECTED TALKS & PRESS

Science x AI Summit 2026 , <i>Speaker</i> (SAIR Foundation & Caltech)	Sep 2026
AI-Native Academia Workshop , NeurIPS 2026, Atlanta (sponsored by ARA Lab)	Dec 2026
Agent-Native Research Artifacts , <i>Invited talk</i> , Lossfunk research community	Aug 2026
Press : IEEE Spectrum (Aug 2026), QbitAI (Aug 2026), Synced (Jun 2026) on Agent-Native Research Artifacts	

COMMUNITY SERVICE

Reviewer , TMC, TPDS, ICDM, MLSys Workshop	
Computer Science Engineering Graduate Student Organization at U of M , <i>DEI Chair</i>	May 2023 - Apr 2025
Student Union of Joint Institute at SJTU , <i>Vice President</i>	Jun 2017 - Aug 2018

RESEARCH EXPERIENCES

ARA: Agent-Native Research Artifacts	
<i>Founder, ARA Commons, Inc.</i>	2026 - Now

- ARA is an executable, verifiable, forkable research package that replaces the narrative PDF paper: scientific logic, executable code, exploration graph, and raw evidence in one machine-readable artifact that agents can reproduce and extend.
- <https://www.agenticresearch.sh/>

Curie: Automated and Rigorous Scientific Experimentation with AI Agents

Advisor: Prof. Mosharaf Chowdhury, Prof. Ang Chen

2024 - 2026

- Curie is the first AI-agent framework designed for automated and rigorous scientific experimentation. Curie helps answer your curiosity through end-to-end experimentation automation, ensuring that every step—from hypothesis formulation to result interpretation—is conducted with precision, reliability, and reproducibility.
- <https://github.com/Just-Curious/Curie>

Large Language Models (LLM) Energy Leaderboard

Advisor: Prof. Mosharaf Chowdhury

2023 - 2024

- Developed an evaluation tool that quantifies the energy consumption of LLMs under different serving scenarios.
- Implemented an online interactive system allowing users to compare the generated content and energy efficiency of different LLMs, thereby understanding trade-offs between performance and energy consumption.
- <https://ml.energy/leaderboard/>

FedScale: A Scalable and Extensible Federated Learning (FL) Benchmark

Advisor: Prof. Mosharaf Chowdhury

2021 - 2022

- Created a platform capable of simulating the behavior of millions of user devices, thereby allowing FL developers to evaluate the performance of their FL applications.
- Collected and formatted the largest benchmarking dataset for various FL tasks, focusing on various challenges like data heterogeneity, device heterogeneity, and connectivity conditions.

High-dimensional Data Index: Adaptive Product Quantization Supporting Data Streaming

Advisor: Prof. Samuel Madden, MIT

May 2019 - Jan 2020

- Proposed an ANN (appropriate nearest neighbor) search model based on product quantization (PQ) to support fast ANN searches in high dimensional scalable dynamic databases with high query speed and high accuracy simultaneously.

COMPUTER SKILLS

Skill: C++, Python, Rust, SQL, Java, Matlab, C, HTML